

EE-102: Electric Circuit Analysis

LECTURE NO 7: *Capacitors and Capacitance*

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

1. Introduction
2. The Electric Field
3. Capacitance
4. Transients in Capacitive Networks: The Charging Phase
5. Transients in Capacitive Networks: The Discharging Phase
6. Capacitors in Parallel and Series
7. Energy stored by a capacitor
8. More Complex Circuits

1. Introduction

- Like the resistor, capacitor is a two-terminal device, but its characteristics are totally different from those of a resistor. In fact, the capacitor displays its true characteristics only when a change in the voltage or current is made in the network.
- All the power delivered to a resistor is dissipated in the form of heat. An ideal capacitor, however, stores the energy delivered to it in a form that can be returned to the system.

2. The Electric Field (1/3)

- Electric field is represented by *electric flux lines*, which are drawn to indicate the strength of the electric field at any point around the charged body. The denser the lines of flux, the stronger is the electric field. In Fig. 1, next slide for example, the electric field strength is stronger in region *a* than region *b* because the flux lines are denser in region *a* than in *b*. That is, the same number of flux lines pass through each region, but the area A_1 is much smaller than area A_2 . The symbol for electric flux is the Greek letter ψ (psi). The flux per unit area (flux density) is represented by the capital letter D and is determined by:

$$D = \frac{\psi}{A} \quad (\text{flux/unit area})$$

2. The Electric Field (2/3)

- The larger the charge Q in coulombs, the greater is the number of flux lines extending or terminating per unit area, independent of the surrounding medium. Twice the charge produces twice the flux per unit area. The two can therefore be equated: $\psi \equiv Q$ (coulombs, C)

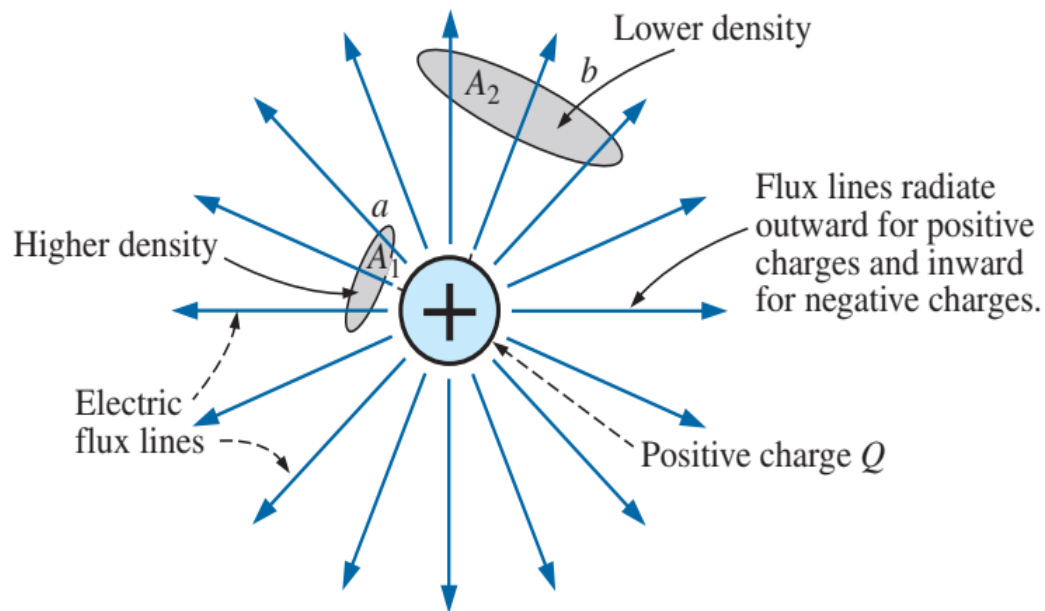


FIG. 1

Flux distribution from an isolated positive charge.

2. The Electric Field (3/3)

- By definition, the electric field strength (designated by the capital script letter $\mathcal{E}$) at a point is the force acting on a unit positive charge at that point; that is, $\mathcal{E} = F / Q_t$ (newtons/coulomb, N/C). Where Q_t is a small positive test charge.
- To illustrate, let us determine the field about the point charge Q of Figure 2. When the test charge is placed near Q , it experiences a force of $F = kQQ_t/r^2$ (Coulomb's law). The constant in Coulomb's law is actually equal to $1/4\pi\epsilon$. Thus, $F = QQ_t/4\pi\epsilon r^2$, and from above Equation

$$\mathcal{E} = \frac{F}{Q_t} = \frac{Q}{4\pi\epsilon r^2} \text{ (N/C)}$$

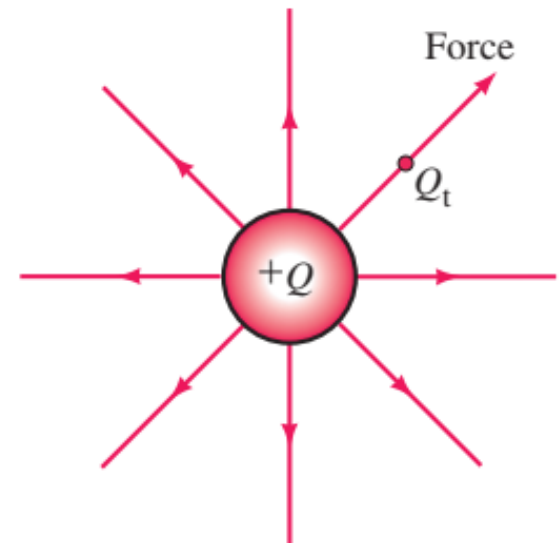
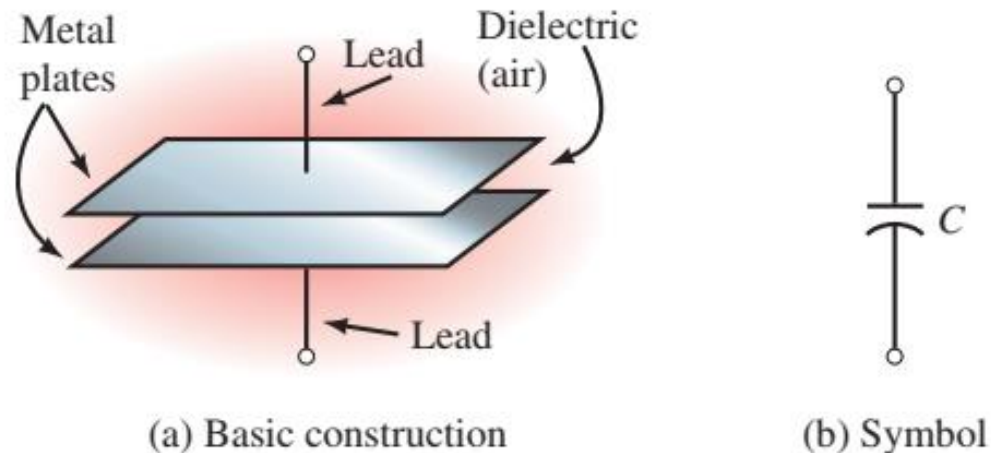


Fig. 2: Force on a test charge Q_t

3. Capacitance (1/4)

- A *capacitor* consists of two conductors separated by an insulator. One of its basic forms is the parallel-plate capacitor shown in Figure 3. It consists of two metal plates separated by a non-conducting material (i.e., an insulator) called a dielectric. The *dielectric* may be air, oil, mica, plastic, ceramic, or other suitable insulating material.



3. Capacitance (2/4)

- Since the plates of the capacitor are metal, they contain huge numbers of free electrons. In their normal state, however, they are uncharged, that is, there is no excess or deficiency of electrons on either plate. If a dc source is now connected (Figure 4, next slide), electrons are pulled from the top plate by the positive potential of the battery and the same number deposited on the bottom plate. This leaves the top plate with a deficiency of electrons (i.e., positive charge) and the bottom plate with an excess (i.e., negative charge). In this state, the capacitor is said to be **charged**. (Note that no current can pass through the dielectric between the plates- thus, the movement of electrons illustrated in Figure 4 will cease when the capacitor reaches full charge.)

3. Capacitance (3/4)

- Capacitance is a measure of a capacitor's ability to store charge on its plates- in other words, its storage capacity.
- In addition, the higher the capacitance of a capacitor, the greater is the amount of charge stored on the plates for the same applied voltage.
- The relationship connecting the applied voltage, the charge on the plates, and the capacitance level is defined by the following equation:

$$C = \frac{Q}{V}$$

C = farads (F)
 Q = coulombs (C)
 V = volts (V)

$$Q = CV$$

(coulombs, C)

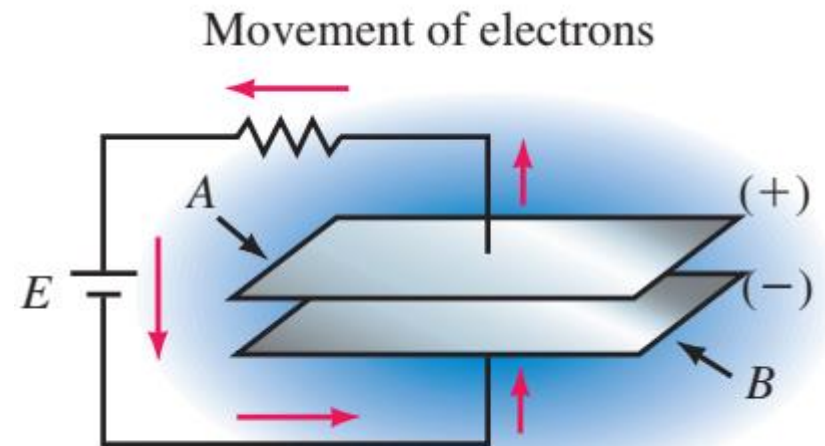


Fig. 4

3. Capacitance (4/4)

EXAMPLE 1

- a. If 82.4×10^{14} electrons are deposited on the negative plate of a capacitor by an applied voltage of 60 V, find the capacitance of the capacitor.
- b. If 40 V are applied across a $470 \mu\text{F}$ capacitor, find the charge on the plates.

Solutions:

- a. First find the number of coulombs of charge as follows:

$$82.4 \times 10^{14} \text{ electrons} \left(\frac{1 \text{ C}}{6.242 \times 10^{18} \text{ electrons}} \right) = 1.32 \text{ mC}$$

and then

$$C = \frac{Q}{V} = \frac{1.32 \text{ mC}}{60 \text{ V}} = \mathbf{22 \mu\text{F}} \quad (\text{a standard value})$$

- b. $Q = CV = (470 \mu\text{F})(40 \text{ V}) = \mathbf{18.8 \text{ mC}}$

4. Transients in Capacitive Networks: The Charging Phase (1/22)

- The placement of charge on the plates of a capacitor does not occur instantaneously. Instead, it occurs over a period of time determined by the components of the network. The charging phase- the phase during which charge is deposited on the plates- can be described by reviewing the response of the simple series circuit in Fig. 4. The circuit has been redrawn in Fig. 5 with the symbol for a fixed capacitor.

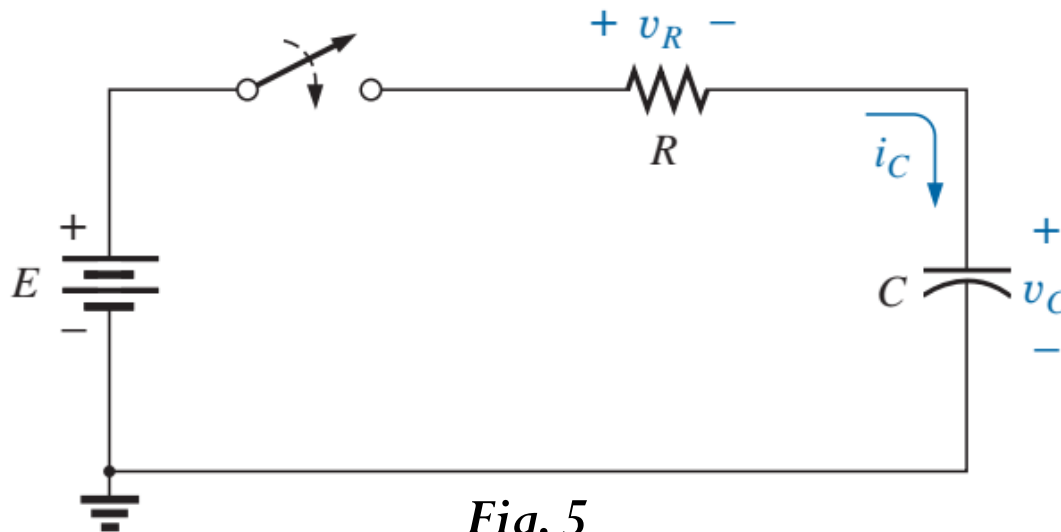


Fig. 5

4. Transients in Capacitive Networks: The Charging Phase (2/22)

- Recall that the instant the switch is closed, electrons are drawn from the top plate and deposited on the bottom plate by the battery, resulting in a net positive charge on the top plate and a negative charge on the bottom plate. The transfer of electrons is very rapid at first, slowing down as the potential across the plates approaches the applied voltage of the battery. Eventually, when the voltage across the capacitor equals the applied voltage, the transfer of electrons ceases, and the plates have a net charge determined by $Q = CV_C = CE$. This period of time during which charge is being deposited on the plates is called the *transient period*- a period of time where the voltage or current changes from one steady-state level to another.

- $Q = CV_C$
- $V_C = E$ (when the capacitor is fully charged.)
- $Q = CE$

4. Transients in Capacitive Networks: The Charging Phase (3/22)

- Since the voltage across the plates is directly related to the charge on the plates by $V = Q/C$, a plot of the voltage across the capacitor will have the same shape as a plot of the charge on the plates over time. As shown in Fig. 6 next slide, the voltage across the capacitor is zero volts when the switch is closed ($t = 0$ s). It then builds up very quickly at first since charge is being deposited at a very high rate of speed. As time passes, the charge is deposited at a slower rate, and the change in voltage drops off. The voltage continues to grow, but at a much slower rate. Eventually, as the voltage across the plates approaches the applied voltage, the charging rate is very slow, until finally the voltage across the plates is equal to the applied voltage- the transient phase has passed.

4. Transients in Capacitive Networks: The Charging Phase (4/22)

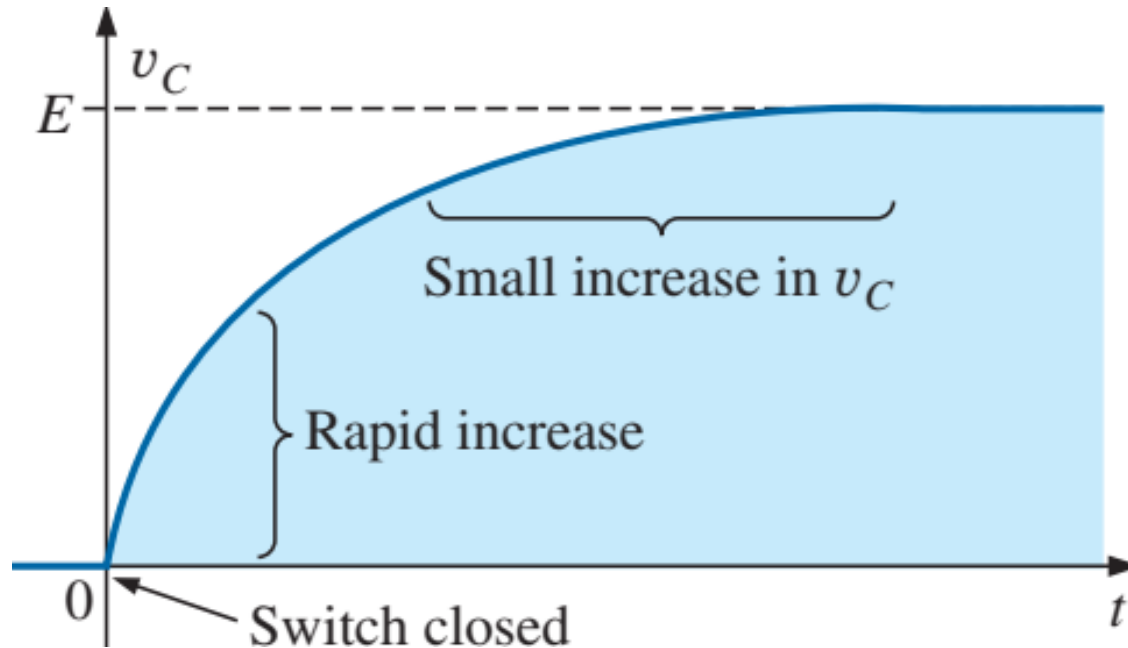


Fig. 6: v_C during charging phase

4. Transients in Capacitive Networks: The Charging Phase (5/22)

- Fortunately, the waveform in Fig. 6 from beginning to end can be described using the mathematical function e^{-x} . It is an exponential function that decreases with time, as shown in Fig. 7 next slide.
- A plot of $1 - e^{-x}$ is also provided in Fig. 7 since it is a component of the voltage v_C in Fig. 7. When e^{-x} is 1, $1 - e^{-x}$ is zero, as shown in Fig. 7, and when e^{-x} decreases in magnitude, $1 - e^{-x}$ approaches 1, as shown in the same figure.
- We simply place the exponential in the proper mathematical form as follows:

$$v_C = E(1 - e^{-t/\tau}) \quad \text{charging} \quad (\text{volts, V}) \quad \dots\dots(1)$$

4. Transients in Capacitive Networks: The Charging Phase (6/22)

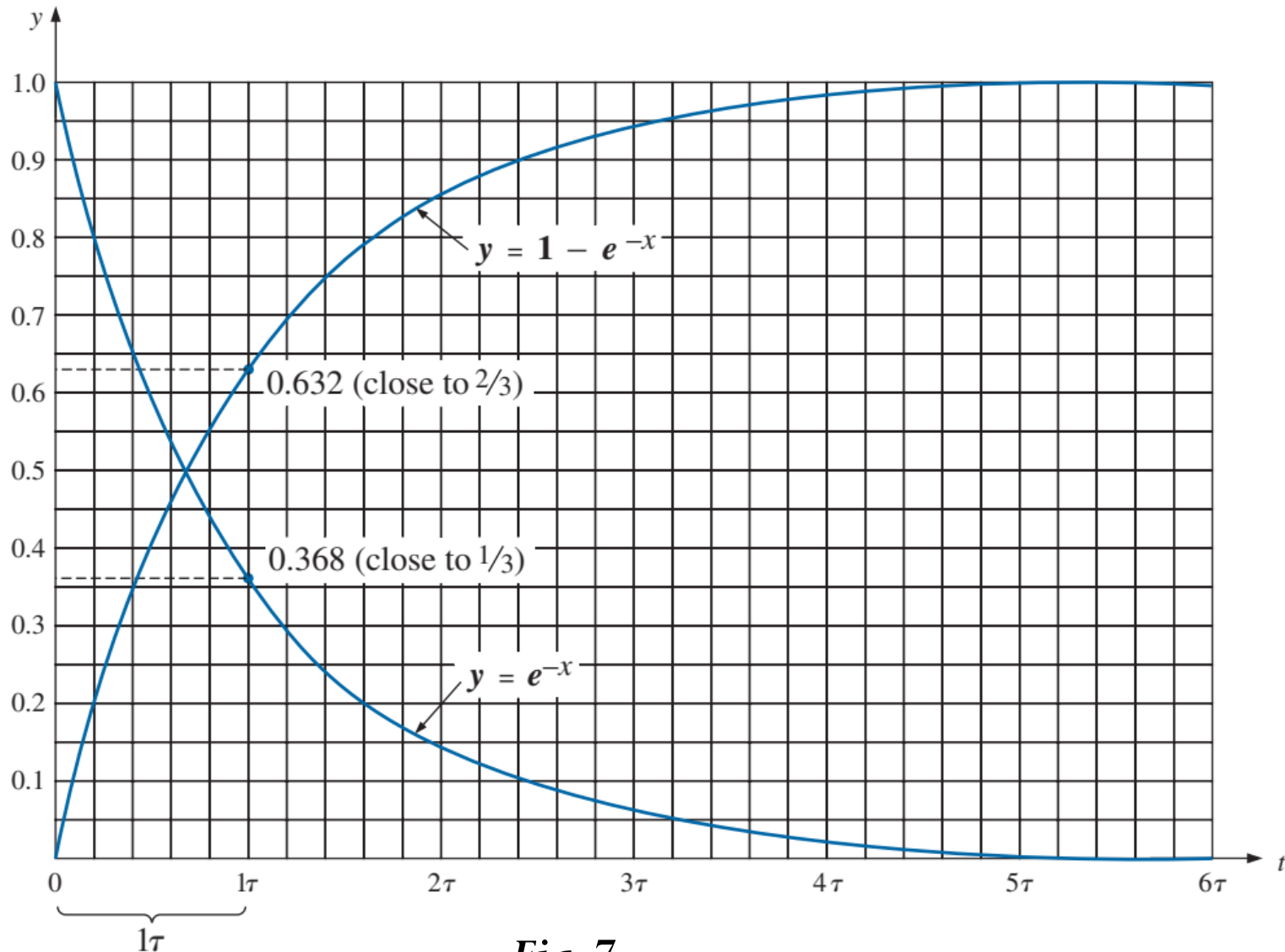


Fig. 7

4. Transients in Capacitive Networks: The Charging Phase (7/22)

- First note in Eq. (1) that the voltage v_C is written in *lowercase* (not capital) point out that it is a function that will change with time- it is not a constant. The exponent of the exponential function is no longer just x , but now is time (t) divided by a constant τ , the Greek letter *tau*. The quantity τ is defined by:

$$\tau = RC \quad (\text{time, s}) \quad \dots\dots(2)$$

The factor τ , called the **time constant** of the network, has the units of

- time, as shown below using some of the basic equations introduced earlier in this text:

$$\tau = RC = \left(\frac{V}{I}\right)\left(\frac{Q}{V}\right) = \left(\frac{\cancel{V}}{Q/t}\right)\left(\frac{Q}{\cancel{V}}\right) = t \text{ (seconds)}$$

- $R = V/I$
- $Q = CV$
- $C = Q/V$
- $I = Q/t$

4. Transients in Capacitive Networks: The Charging Phase (8/22)

- A plot of Eq. (1) results in the curve in Fig. 8, whose shape is an exact match with that in Fig. 6 in slide 15.

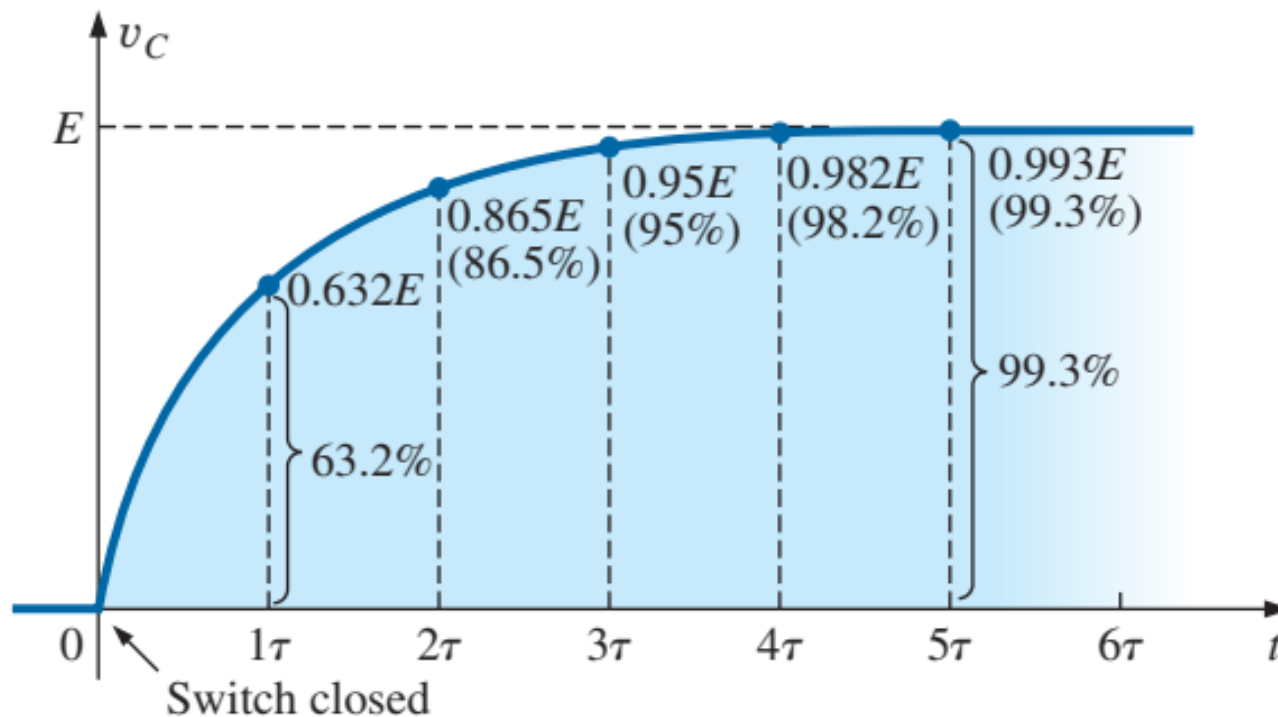


Fig. 8

4. Transients in Capacitive Networks: The Charging Phase (9/22)

- In Eq. (1), if we substitute $t = 0$ s, we find that

$$e^{-t/\tau} = e^{-0/\tau} = e^{-0} = \frac{1}{e^0} = \frac{1}{1} = 1$$

and

$$v_C = E(1 - e^{-t/\tau}) = E(1 - 1) = \mathbf{0\ V}$$

- For $t = 1\tau$ in Eq. (1), results in:

$$e^{-t/\tau} = e^{-\tau/\tau} = e^{-1} \cong 0.368$$

and

$$v_C = E(1 - e^{-t/\tau}) = E(1 - 0.368) = \mathbf{0.632E}$$

4. Transients in Capacitive Networks: The Charging Phase (10/22)

- For $t = 5\tau$ in Eq. (1), results in:

$$e^{-t/\tau} = e^{-5\tau/\tau} = e^{-5} \cong 0.007$$

and

$$v_C = E(1 - e^{-t/\tau}) = E(1 - 0.007) = \mathbf{0.993E} \cong E$$

- In fact, we can conclude from the results just obtained that:
 - *The voltage across a capacitor in a dc network is essentially equal to the applied voltage after five time constants of the charging phase have passed.*
- Or in more general terms:
 - *The transient or charging phase of a capacitor has essentially ended after five time constants.*

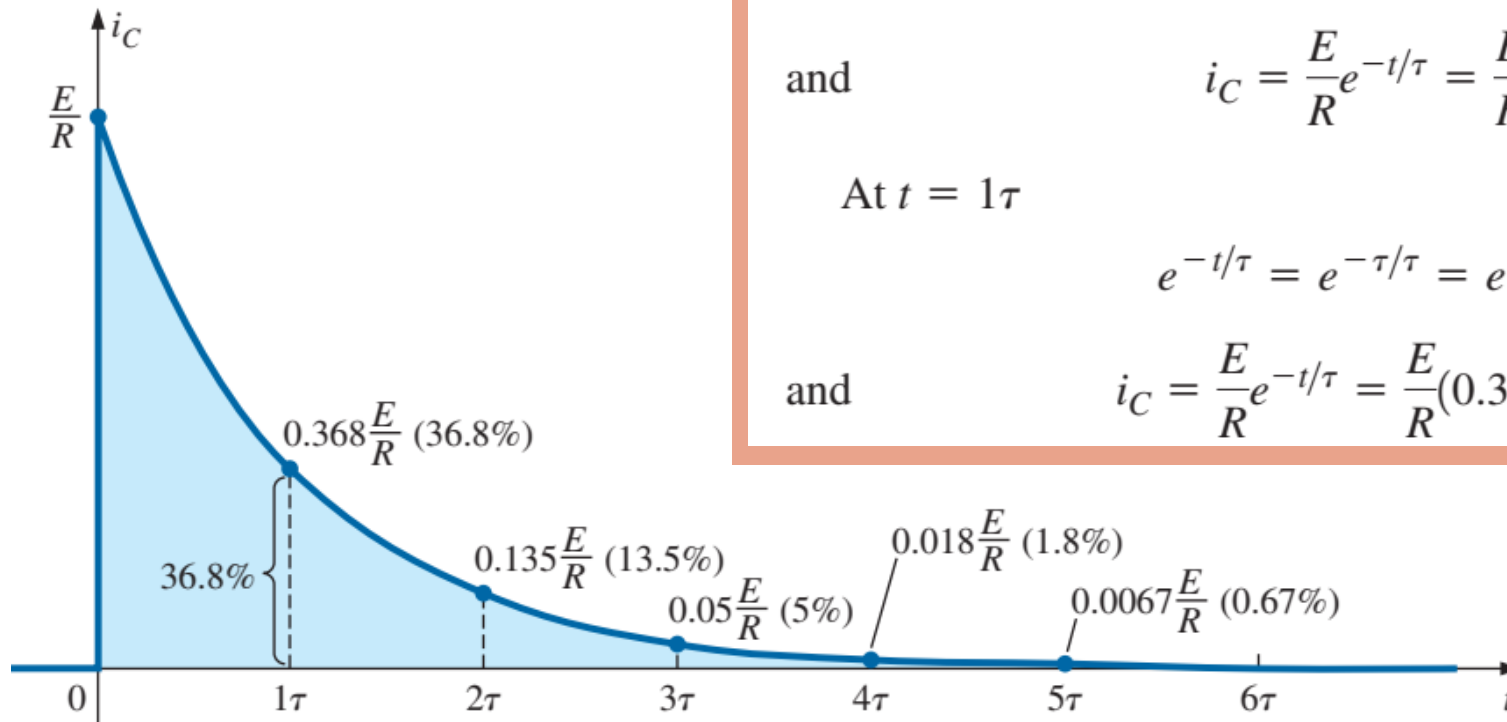
4. Transients in Capacitive Networks: The Charging Phase (11/22)

- It is indeed fortunate that the same exponential function can be used to plot the current of the capacitor versus time. When the switch is first closed, the flow of charge or current jumps very quickly to a value limited by the applied voltage and the circuit resistance, as shown in Fig. 9 next slide. The rate of deposit, and hence the current, then decreases quite rapidly, until eventually charge is not being deposited on the plates and the current drops to zero amperes. The equation for the current is:

$$i_C = \frac{E}{R} e^{-t/\tau} \quad \text{(amperes, A)} \quad \dots\dots(3)$$

charging

4. Transients in Capacitive Networks: The Charging Phase (12/22)



At $t = 0$ s

$$e^{-t/\tau} = e^{-0} = 1$$

and

$$i_C = \frac{E}{R} e^{-t/\tau} = \frac{E}{R} (1) = \frac{E}{R}$$

At $t = 1\tau$

$$e^{-t/\tau} = e^{-\tau/\tau} = e^{-1} \cong 0.368$$

and

$$i_C = \frac{E}{R} e^{-t/\tau} = \frac{E}{R} (0.368) = \mathbf{0.368 \frac{E}{R}}$$

Fig. 9

4. Transients in Capacitive Networks: The Charging Phase (13/22)

- In general, Fig.9 clearly reveals that:
 - *The current of a capacitive dc network is essentially zero amperes after five time constants of the charging phase have passed.*
- It is also important to recognize that
 - *During the charging phase, the major change in voltage and current occurs during the first time constant.*
- Returning to Figs. 8 and 9, note that when the voltage across the capacitor reaches the applied voltage E , the current drops to zero amperes, as reviewed in Fig. 10 next slide. These conditions match those of an open circuit, permitting the following conclusion:
 - *A capacitor can be replaced by an open-circuit equivalent once the charging phase in a dc network has passed.*

4. Transients in Capacitive Networks: The Charging Phase (14/22)

- As shown in figure 11 given below:
 - A capacitor has the characteristics of a short-circuit equivalent at the instant the switch is closed in an uncharged series R-C circuit.*

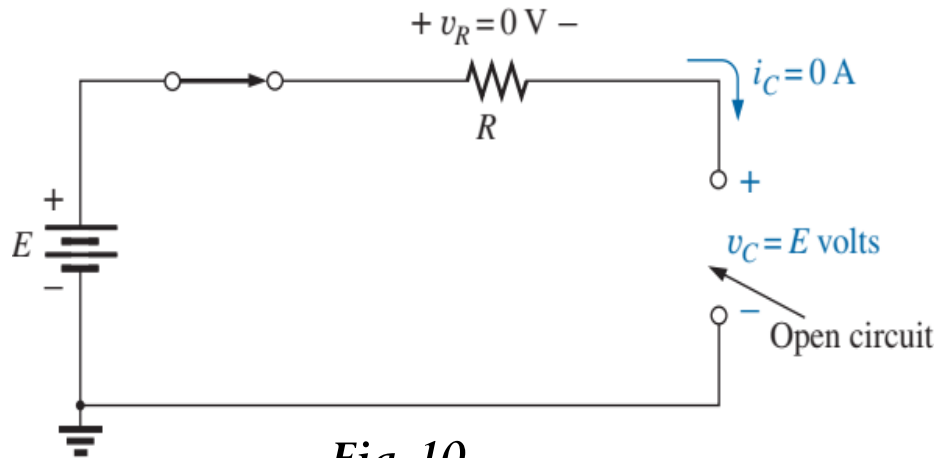


Fig. 10

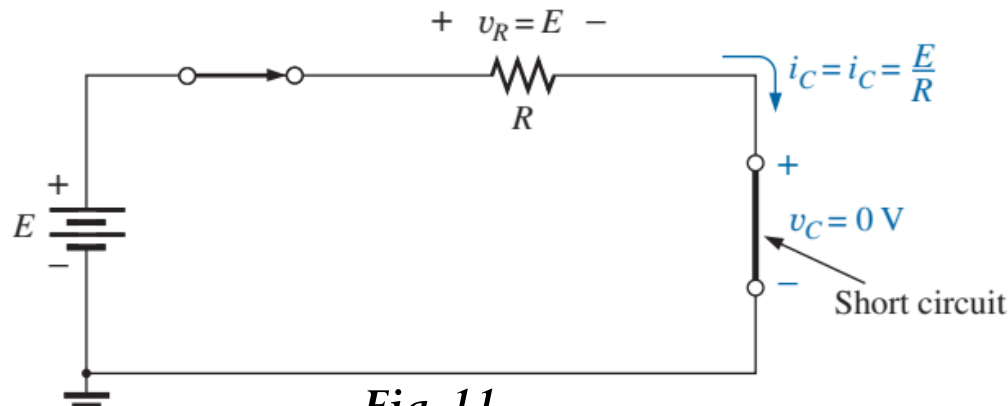


Fig. 11

4. Transients in Capacitive Networks: The Charging Phase (15/22)

- The voltage across a capacitor cannot change instantaneously.
- Since the resistor and the capacitor in Fig. 5 are in series, the current through the resistor is the same as that associated with the capacitor. The voltage across the resistor can be determined by using Ohm's law in the following manner:

$$v_R = i_R R = i_C R$$

so that

$$v_R = \left(\frac{E}{R} e^{-t/\tau} \right) R$$

and

$$\boxed{v_R = E e^{-t/\tau}} \text{ charging} \quad (\text{volts, V}) \dots\dots(4)$$

4. Transients in Capacitive Networks: The Charging Phase (16/22)

- A plot of the voltage as shown in Fig. 12 has the same shape as that for the current because they are related by the constant R .

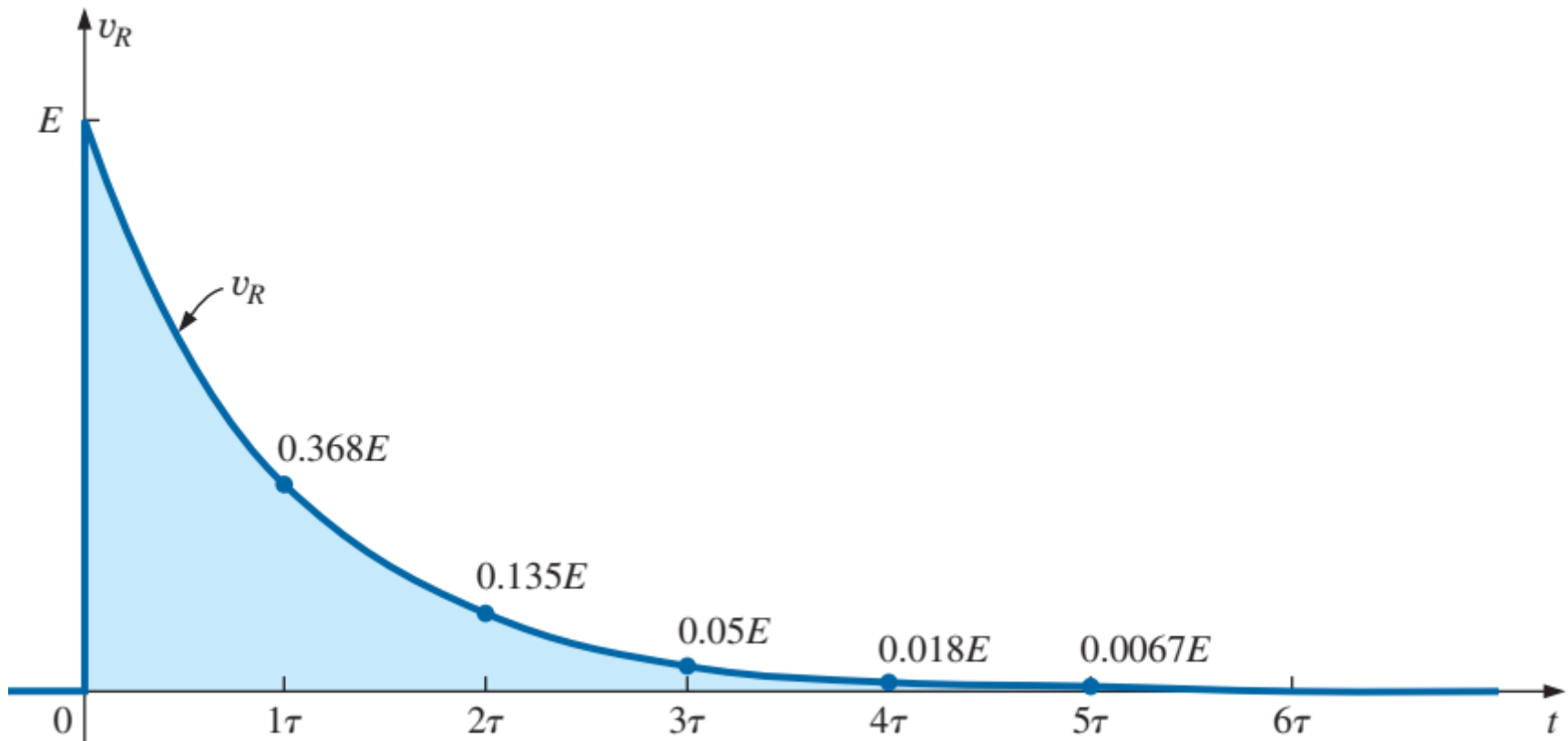


Fig. 12

4. Transients in Capacitive Networks: The Charging Phase (17/22)

- Capacitor with an Initial Voltage

- Suppose a previously charged capacitor has not been discharged and thus still has voltage on it. Let this voltage be denoted as V_0 . (V_0 is referred to as an *initial condition*.) If the capacitor is now placed in a circuit like that in Figure 5, the voltage and current during charging will be affected by the initial voltage. In this case, Equations 1 and 3 become:

$$v_C = E + (V_0 - E)e^{-t/\tau} \quad \dots\dots(5)$$

$$i_C = \frac{E - V_0}{R} e^{-t/\tau} \quad \dots\dots(6)$$

4. Transients in Capacitive Networks: The Charging Phase (18/22)

Example 1: For the circuit in Figure 13 given below:

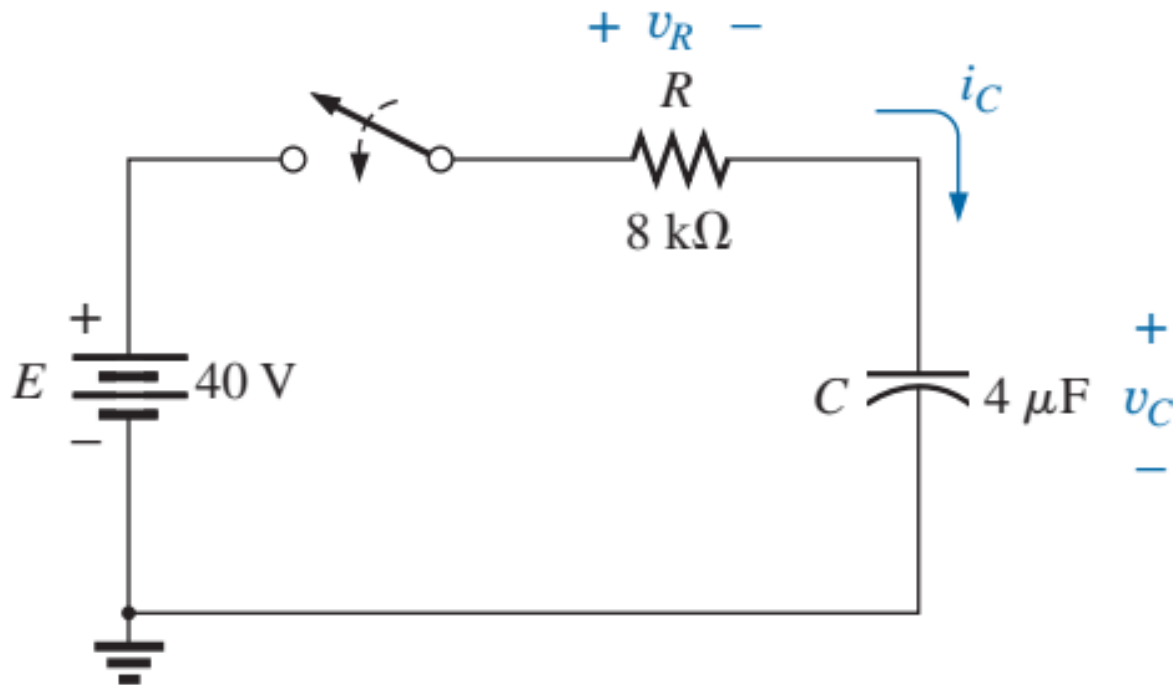


Fig. 13

4. Transients in Capacitive Networks: The Charging Phase (19/22)

Example 1: (continued...)

- Find the mathematical expression for the transient behavior of v_C , i_C , and v_R if the switch is closed at $t = 0$ s.
- Plot the waveform of v_C versus the time constant of the network.
- Plot the waveform of v_C versus time.
- Plot the waveforms of i_C and v_R versus the time constant of the network.
- What is the value of v_C at $t = 20$ ms?
- On a practical basis, how much time must pass before we can assume that the charging phase has passed?
- When the charging phase has passed, how much charge is sitting on the plates?
- If the capacitor has a leakage resistance of $10,000 \text{ M}\Omega$, what is the initial leakage current? Once the capacitor is separated from the circuit, how long will it take to totally discharge, assuming a linear (unchanging) discharge rate?

4. Transients in Capacitive Networks: The Charging Phase (20/22)

Solution:

a. The time constant of the network is

$$\tau = RC = (8 \text{ k}\Omega)(4 \text{ }\mu\text{F}) = 32 \text{ ms}$$

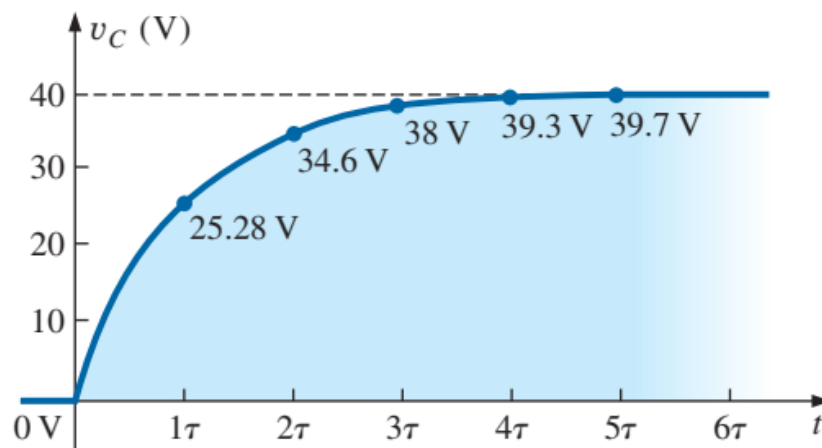
resulting in the following mathematical equations:

$$v_C = E(1 - e^{-t/\tau}) = 40 \text{ V}(1 - e^{-t/32\text{ms}})$$

$$i_C = \frac{E}{R}e^{-t/\tau} = \frac{40 \text{ V}}{8 \text{ k}\Omega}e^{-t/32\text{ms}} = 5 \text{ mA}e^{-t/32\text{ms}}$$

$$v_R = Ee^{-t/\tau} = 40 \text{ V}e^{-t/32\text{ms}}$$

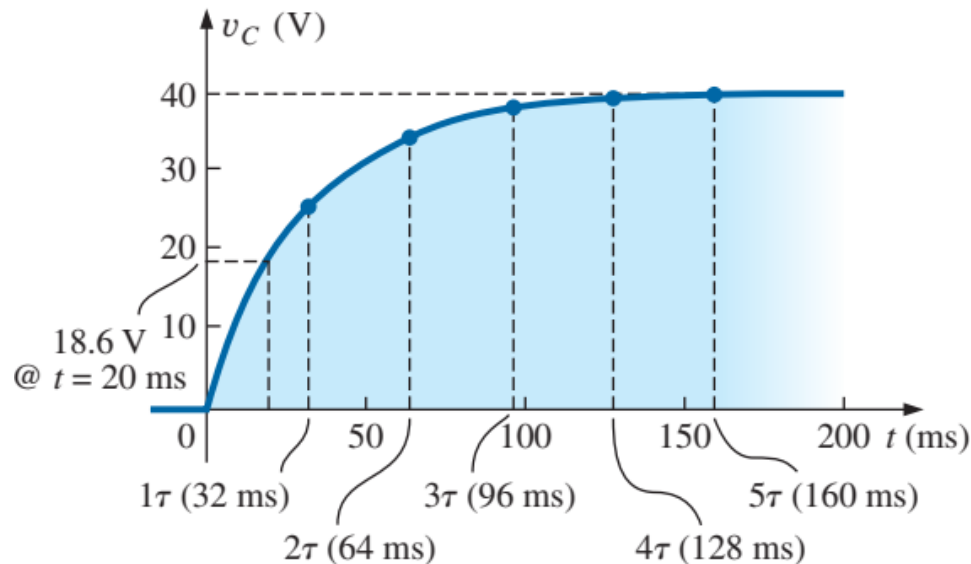
b. The resulting plot



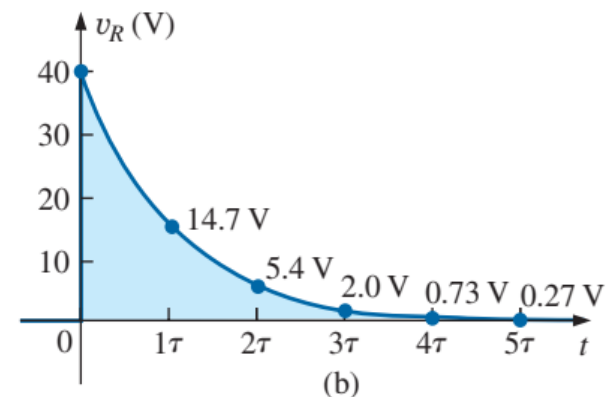
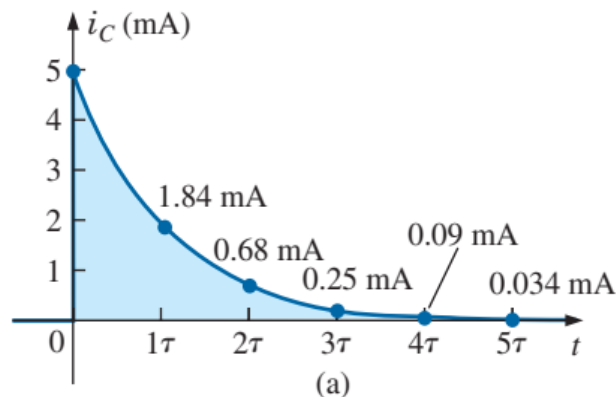
4. Transients in Capacitive Networks: The Charging Phase (21/22)

Solution:

- c. The horizontal scale will now be against time rather than time constants



- d. Both plots appear



4. Transients in Capacitive Networks: The Charging Phase (22/22)

Solution:

- e. Substituting the time $t = 20$ ms results in the following for the exponential part of the equation:

$$e^{t/\tau} = e^{-20 \text{ ms}/32 \text{ ms}} = e^{-0.625} = 0.535$$

$$\begin{aligned}\text{so that } v_C &= 40 \text{ V}(1 - e^{t/32 \text{ ms}}) = 40 \text{ V}(1 - 0.535) \\ &= (40 \text{ V})(0.465) = \mathbf{18.6 \text{ V}}\end{aligned}$$

- f. Assuming a full charge in five time constants results in

$$5\tau = 5(32 \text{ ms}) = \mathbf{160 \text{ ms} = 0.16 \text{ s}}$$

g. $Q = CV = (4 \mu\text{F})(40 \text{ V}) = \mathbf{160 \mu\text{C}}$

- h. Using Ohm's law gives

$$I_{\text{leakage}} = \frac{40 \text{ V}}{10,000 \text{ M}\Omega} = 4 \text{ nA}$$

Finally, the basic equation $I = Q/t$ results in

$$t = \frac{Q}{I} = \frac{160 \mu\text{C}}{4 \text{ nA}} = (40,000 \text{ s}) \left(\frac{1 \cancel{\text{min}}}{60 \text{ s}} \right) \left(\frac{1 \text{ h}}{60 \cancel{\text{min}}} \right) = \mathbf{11.11 \text{ h}}$$

5. Transients in Capacitive Networks: The Discharging Phase (1/8)

- In Fig. 14(a) next slide, a second contact for the switch was added to the circuit in Fig. 5(slide 12) to permit a controlled discharge of the capacitor. With the switch in position 1, we have the charging network described in the last section. Following the full charging phase, if we move the switch to position 2, the capacitor can be discharged through the resulting circuit in Fig. 14(b) next slide. Initially, the current jumps to a relatively high value; then it begins to drop. It drops with time because charge is leaving the plates of the capacitor, which in turn reduces the voltage across the capacitor and thereby the voltage across the resistor and the resulting current.

5. Transients in Capacitive Networks: The Discharging Phase (2/8)

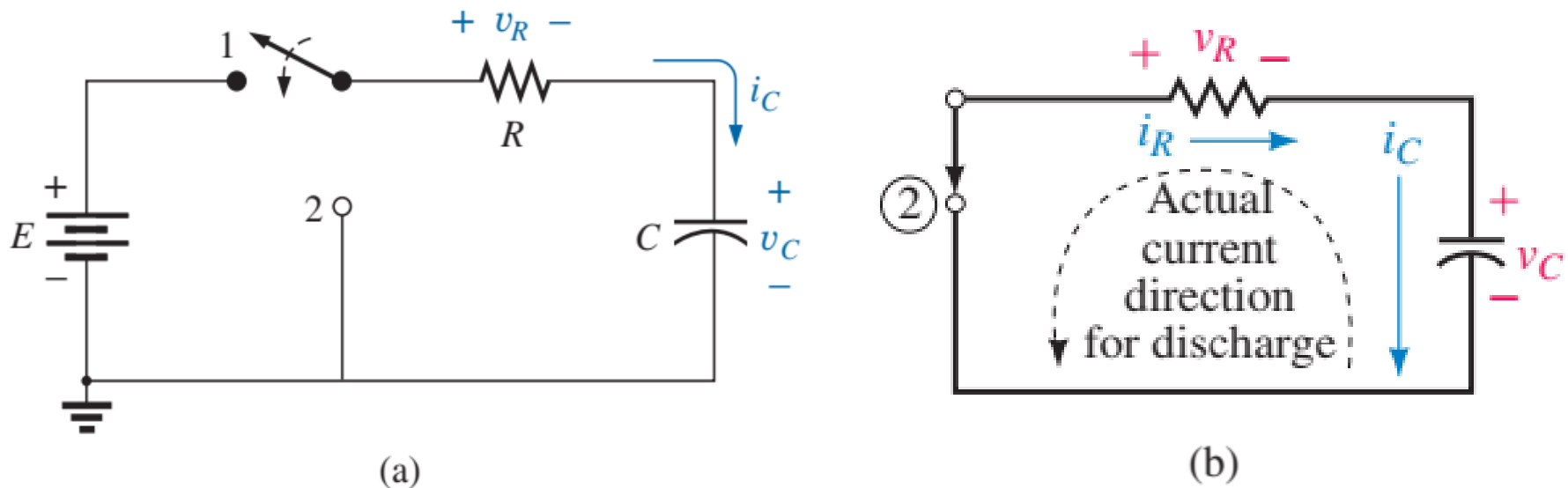


Fig. 14

5. Transients in Capacitive Networks: The Discharging Phase (3/8)

- For the voltage across the capacitor that is decreasing with time, the mathematical expression is:

$$\boxed{v_C = Ee^{-t/\tau}}_{\text{discharging}} \dots\dots(7)$$

- For this circuit, the time constant τ is defined by the same equation as used for the charging phase. That is:

$$\boxed{\tau = RC}_{\text{discharging}} \dots\dots(8)$$

5. Transients in Capacitive Networks: The Discharging Phase (4/8)

- Since the current decreases with time, it will have a similar format:

$$i_C = -\frac{E}{R}e^{-t/\tau} \quad \text{.....(9)}$$

discharging

For the configuration in Fig. 14(b), apply KVL in clockwise direction:

- $-v_C - v_R = 0$
- $-v_C = v_R$
- $v_R = -v_C$

$$v_R = -Ee^{-t/\tau} \quad \text{discharging} \quad \text{.....(10)}$$

5. Transients in Capacitive Networks: The Discharging Phase (5/8)

The complete discharge will occur, for all practical purposes, in five time constants. If the switch is moved between terminals 1 and 2 every five time constants, the wave shapes in Fig. 15 will result for v_C , i_C , and v_R .

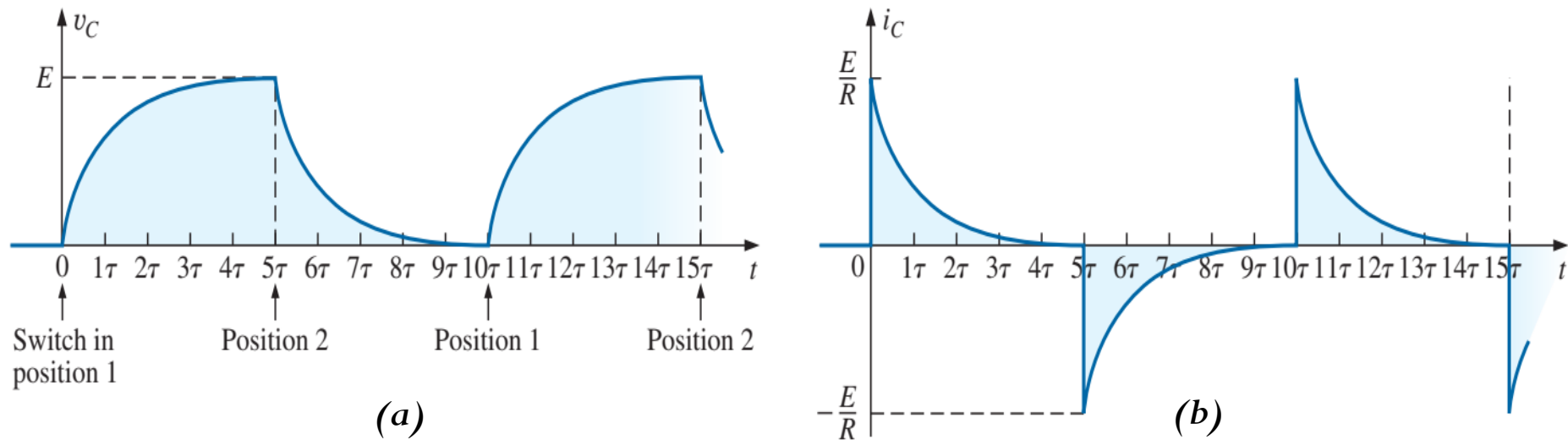
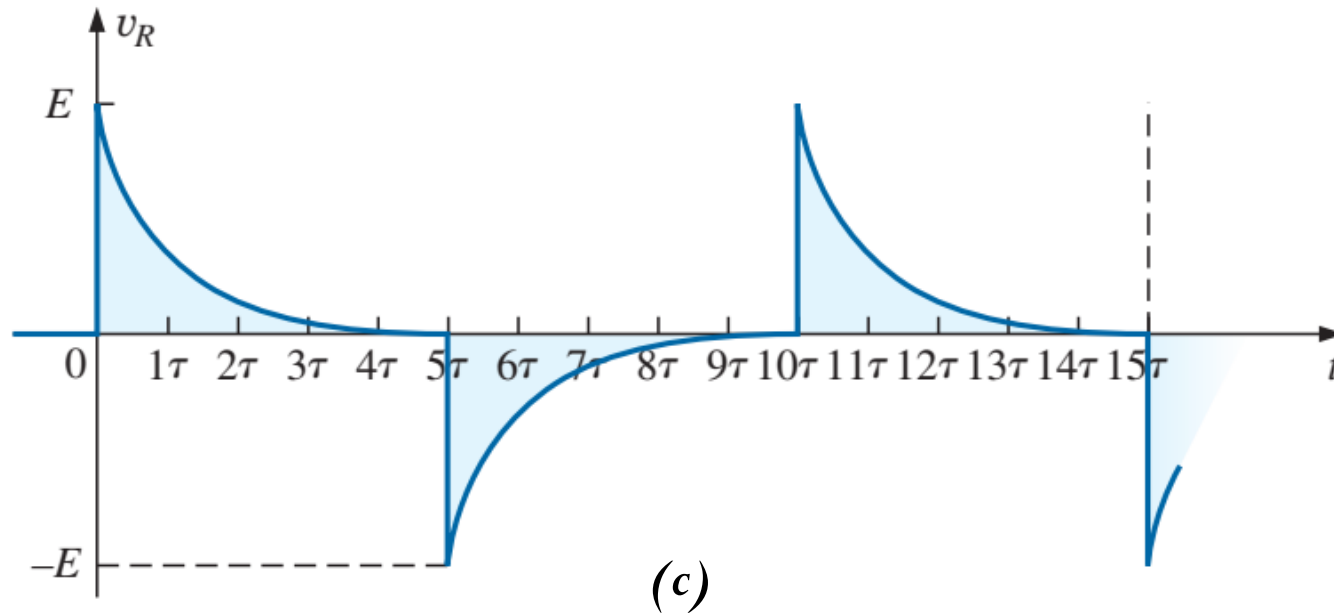


Fig. 15.

5. Transients in Capacitive Networks: The Discharging Phase (6/8)



(c)
Fig. 15

5. Transients in Capacitive Networks: The Discharging Phase (7/8)

EXAMPLE 2 Using the values in Example 1, plot the waveforms for v_C and i_C resulting from switching between contacts 1 and 2 in Fig.14 every five time constants.

Solution: The time constant is the same for the charging and discharging phases. That is,

$$\tau = RC = (8 \text{ k}\Omega)(4 \text{ }\mu\text{F}) = 32 \text{ ms}$$

For the discharge phase, the equations are

$$v_C = Ee^{-t/\tau} = \mathbf{40 \text{ V}e^{-t/32\text{ms}}}$$

$$i_C = -\frac{E}{R}e^{-t/\tau} = \frac{40 \text{ V}}{8 \text{ k}\Omega}e^{-t/32\text{ms}} = \mathbf{-5 \text{ mA}e^{-t/32\text{ms}}}$$

$$v_R = -v_C = \mathbf{-40 \text{ V}e^{-t/32\text{ms}}}$$

5. Transients in Capacitive Networks: The Discharging Phase (8/8)

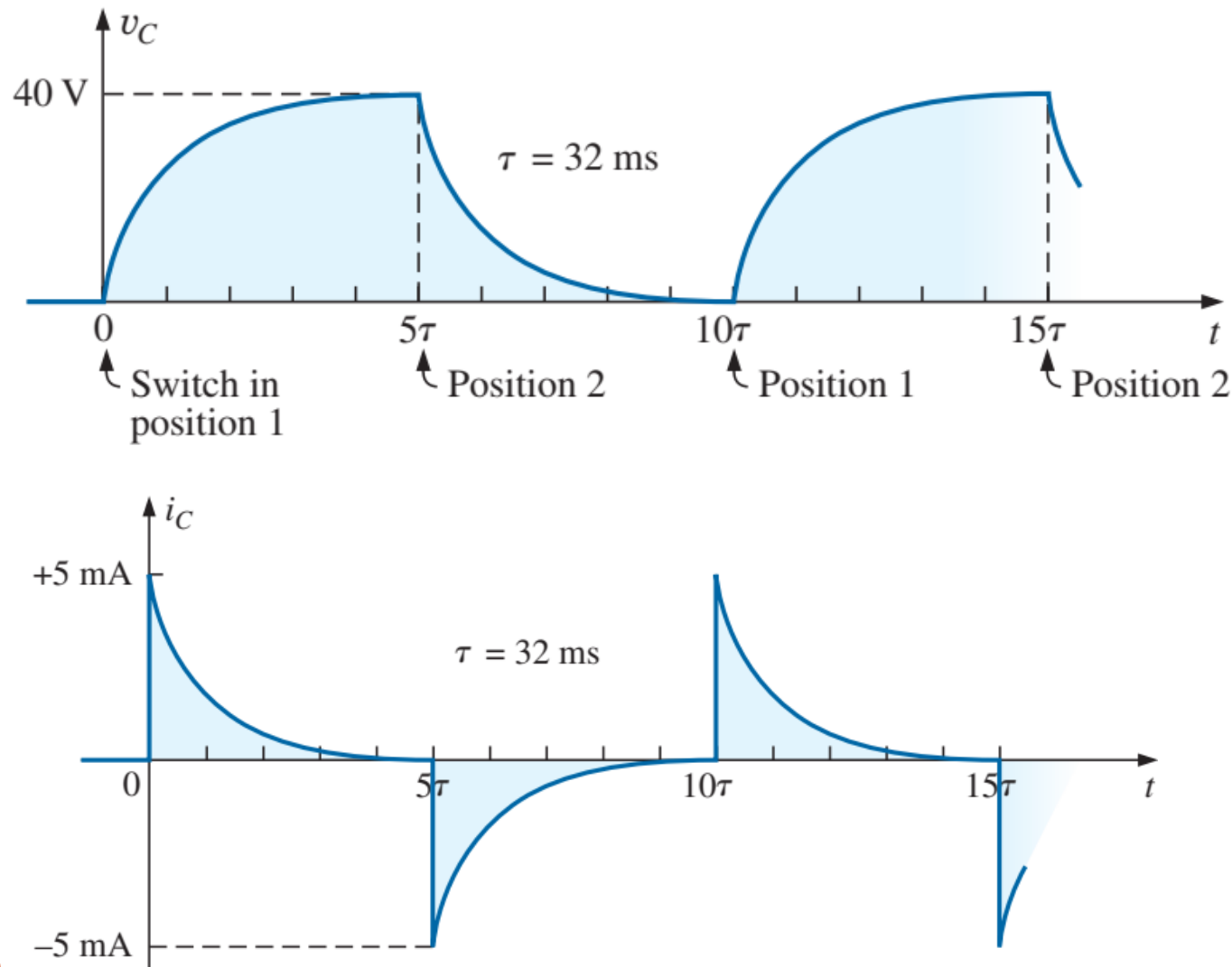


Fig. 16

6. Capacitors in Parallel and Series (1/6)

Capacitors in Parallel

- Consider Figure 17. The charge on capacitor C_1 is given as $Q_1 = C_1 V$. The charge on capacitor C_2 is given as $Q_2 = C_2 V$.
- Thus total charge can be found as :
 - $Q_T = Q_1 + Q_2 = C_1 V + C_2 V = (C_1 + C_2) V = (C_T) V$
- Thus total capacitance will be: $C_T = C_1 + C_2 + \dots + C_N$

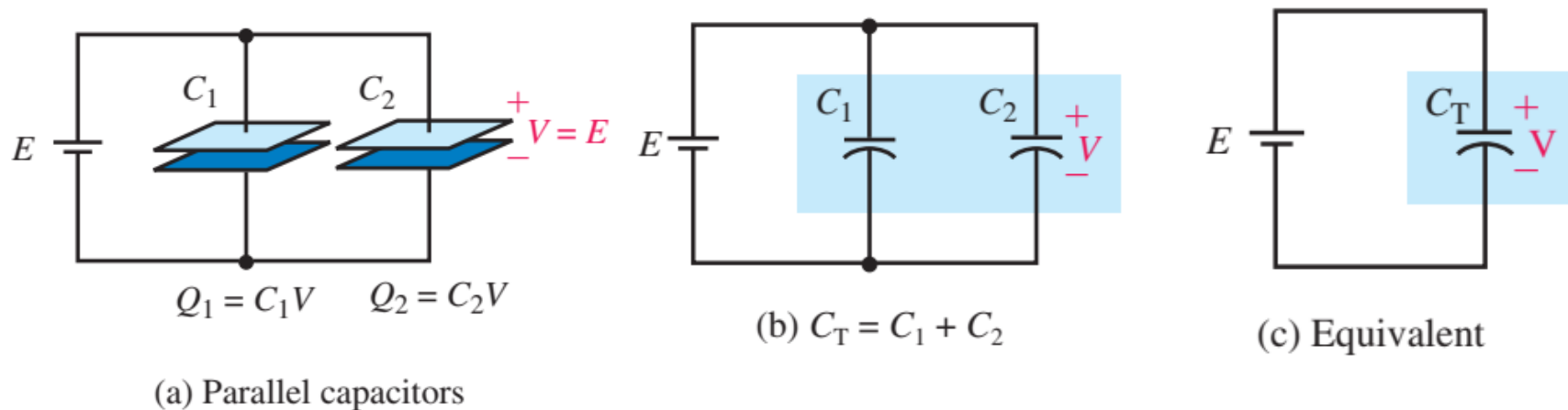


Fig. 17

6. Capacitors in Parallel and Series (2/6)

Example 3

A 10- μF , a 15- μF , and a 100- μF capacitor are connected in parallel across a 50-V source. Determine the following:

- Total capacitance.
- Total charge stored.
- Charge on each capacitor.

Solution

a. $C_T = C_1 + C_2 + C_3 = 10\ \mu\text{F} + 15\ \mu\text{F} + 100\ \mu\text{F} = 125\ \mu\text{F}$

b. $Q_T = C_T V = (125\ \mu\text{F})(50\ \text{V}) = 6.25\ \text{mC}$

c. $Q_1 = C_1 V = (10\ \mu\text{F})(50\ \text{V}) = 0.5\ \text{mC}$

$$Q_2 = C_2 V = (15\ \mu\text{F})(50\ \text{V}) = 0.75\ \text{mC}$$

$$Q_3 = C_3 V = (100\ \mu\text{F})(50\ \text{V}) = 5.0\ \text{mC}$$

$$\text{Check: } Q_T = Q_1 + Q_2 + Q_3 = (0.5 + 0.75 + 5.0)\ \text{mC} = 6.25\ \text{mC}.$$

6. Capacitors in Parallel and Series (3/6)

Capacitors in Series

- Consider Figure 18. The same charge appears on each capacitor. Thus $Q = C_1 V_1$, $Q = C_2 V_2$ and so on.
- Solving for voltages yields $V_1 = Q/C_1$, $V_2 = Q/C_2$ and so on. Applying KVL we get, $V = V_1 + V_2 + \dots + V_N$. Therefore,

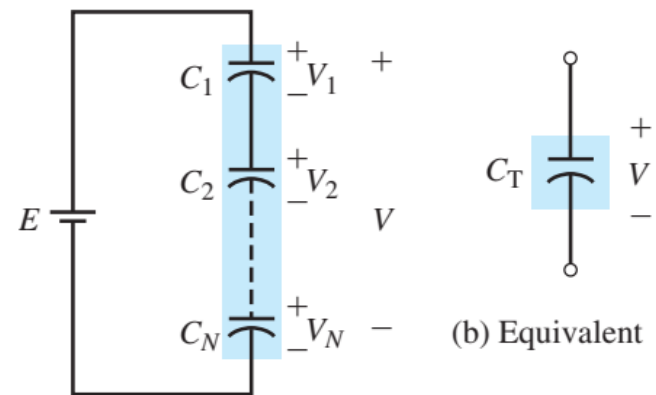
$$V = \frac{Q}{C_1} + \frac{Q}{C_2} + \dots + \frac{Q}{C_N} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \right)$$

But $V = Q/C_T$. Equating this with the right side and cancelling Q yields

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

For two capacitors in series, this reduces to

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$



(a) Series connection **Fig. 18**

6. Capacitors in Parallel and Series (4/6)

EXAMPLE 4 Refer to Figure 19:

- Determine C_T .
- If 50 V is applied across the capacitors, determine Q .
- Determine the voltage on each capacitor.

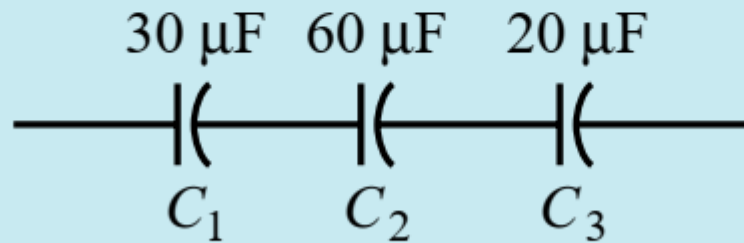


Fig. 19

6. Capacitors in Parallel and Series (5/6)

Solution

$$\begin{aligned}\text{a. } \frac{1}{C_T} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{30 \mu\text{F}} + \frac{1}{60 \mu\text{F}} + \frac{1}{20 \mu\text{F}} \\ &= 0.0333 \times 10^6 + 0.0167 \times 10^6 + 0.05 \times 10^6 = 0.1 \times 10^6\end{aligned}$$

Therefore as indicated in (b),

$$C_T = \frac{1}{0.1 \times 10^6} = 10 \mu\text{F}$$

$$\text{b. } Q = C_T V = (10 \times 10^{-6} \text{ F})(50 \text{ V}) = 0.5 \text{ mC}$$

$$\text{c. } V_1 = Q/C_1 = (0.5 \times 10^{-3} \text{ C})/(30 \times 10^{-6} \text{ F}) = 16.7 \text{ V}$$

$$V_2 = Q/C_2 = (0.5 \times 10^{-3} \text{ C})/(60 \times 10^{-6} \text{ F}) = 8.3 \text{ V}$$

$$V_3 = Q/C_3 = (0.5 \times 10^{-3} \text{ C})/(20 \times 10^{-6} \text{ F}) = 25.0 \text{ V}$$

$$\text{Check: } V_1 + V_2 + V_3 = 16.7 + 8.3 + 25 = 50 \text{ V.}$$

6. Capacitors in Parallel and Series (6/6)

Voltage Divider Rule for Series Capacitors

- Consider Figure 20.

- $Q_T = Q_1$ (As charge of each capacitor is same in series circuit.)
- $Q_T = C_1 V_1$
- $V_1 = Q_T / C_1$
- $V_1 = 1 / C_1 \times (C_T V_T)$
- $V_1 = (C_T / C_1) \times V_T$
- $V_x = (C_T / C_x) \times V_T$

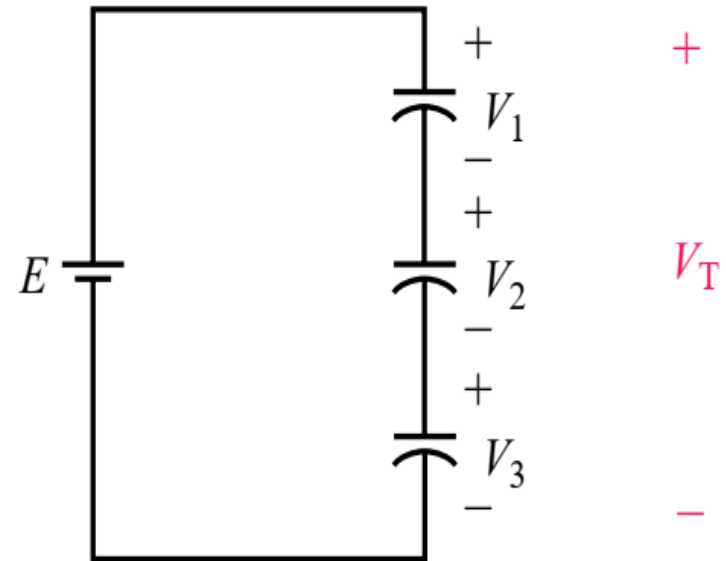


Fig. 20

7. Energy Stored by a Capacitor (1/5)

- An ideal capacitor does not dissipate any of the energy supplied to it. It stores the energy in the form of an electric field between the conducting surfaces. A plot of the voltage, current, and power to a capacitor during the charging phase is shown in Fig. 21. *The energy stored is represented by the shaded area under the power curve.*

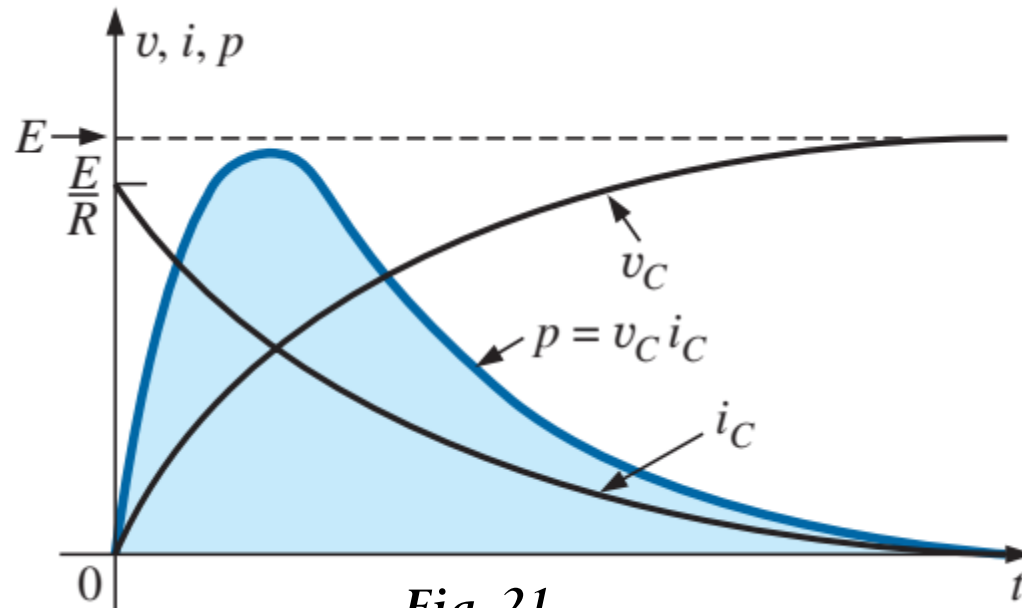


Fig. 21

7. Energy Stored by a Capacitor (2/5)

- Using calculus, we can determine the area under the curve:

$$W_C = \frac{1}{2} CE^2$$

In general,

$$W_C = \frac{1}{2} CV^2 \quad (J)$$

- $W_C = \frac{1}{2} QV$
- As $Q = CV$
- $W_C = \frac{1}{2} (CV) \cdot V$
- $W_C = \frac{1}{2} CV^2$

where V is the steady-state voltage across the capacitor. In terms of Q and C ,

$$W_C = \frac{1}{2} C \left(\frac{Q}{C} \right)^2$$

or

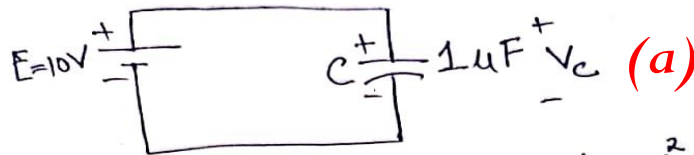
$$W_C = \frac{Q^2}{2C} \quad (J)$$

7. Energy Stored by a Capacitor (3/5)

Example no 5:

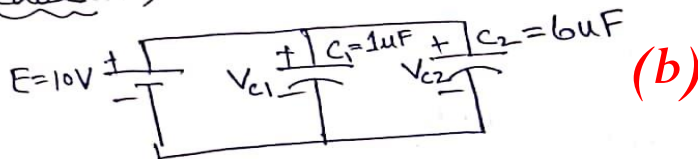
Examples (Related to Energy stored by a capacitor) (1)

Case (i)



$$\text{Energy stored by capacitor} = W_c = \frac{1}{2} C V_c^2 = (1\mu F)(10)^2 \times \frac{1}{2}$$
$$W_c = 50\mu\text{Joules}$$

Case (ii)



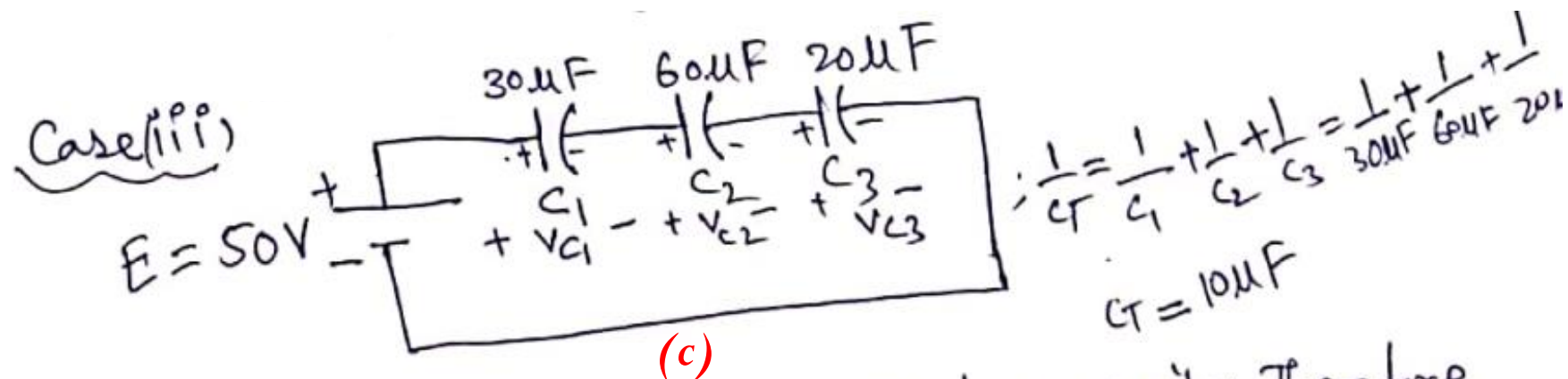
$$\text{Energy stored by a capacitor } C_1: W_{c1} = \frac{1}{2} C_1 \times E^2$$

$$W_{c1} = \frac{1}{2} (1\mu F)(10V)^2 = 50\mu\text{Joules}$$

$$\text{Energy stored by a capacitor } C_2: W_{c2} = \frac{1}{2} C_2 \times E^2$$

$$W_{c2} = \frac{1}{2} (6\mu F)(10V)^2 = 300\mu\text{Joules}$$

7. Energy Stored by a Capacitor (4/5)



* The same charge appears on each capacitor. Therefore

$$Q = C_T V = 10\mu\text{F} \times 50\text{V} = 0.5\text{mC}$$

$$\text{Now } V_{C1} = \frac{Q}{C_1} = \frac{0.5\text{mC}}{30\mu\text{F}} = 16.7\text{V}; V_{C2} = \frac{Q}{C_2} = \frac{0.5\text{mC}}{60\mu\text{F}} = 8.3\text{V}$$

7. Energy Stored by a Capacitor (5/5)

$$V_{c3} = \frac{Q}{C_3} = \frac{0.5mC}{20\mu F} = 25V$$

(2)

Now Energy stored by capacitor C_1

$$W_{c1} = \frac{1}{2} C_1 V_{c1}^2 = \frac{1}{2} (30\mu F) (16.7)^2 = 4.18mJ$$

You can also use formula $W_c = \frac{1}{2} Q V_{c1}$; it give the same solution as W_{c1}

Energy stored by capacitor C_2

$$W_{c2} = \frac{1}{2} C_2 V_{c2}^2 = \frac{1}{2} (60\mu F) (8.3V)^2$$

$$W_{c2} = 2.1mJ$$

Energy stored by capacitor C_3

$$W_{c3} = \frac{1}{2} C_3 V_{c3}^2 = \frac{1}{2} (20\mu F) (25V)^2$$

$$W_{c3} = 6.25mJ$$

8. More Complex Circuits (1/15)

- The charge and discharge equations and the universal time constant curves apply only to circuits of the forms shown in Figures 23 and 24. Fortunately, many circuits can be reduced to these forms using standard circuit reduction techniques such as series and parallel combinations, source conversions, Thévenin's theorem, and so on. Once a circuit has been reduced to its series equivalent, you can use any of the techniques that we have developed so far.

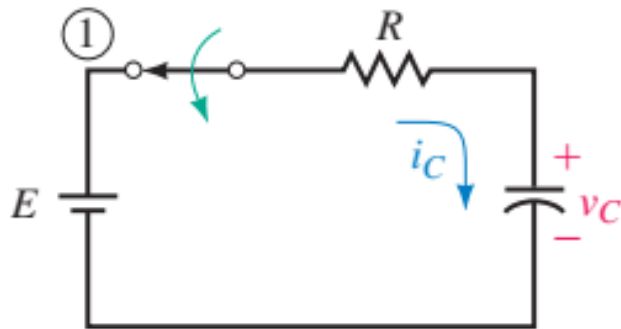


Fig. 23: Charging case

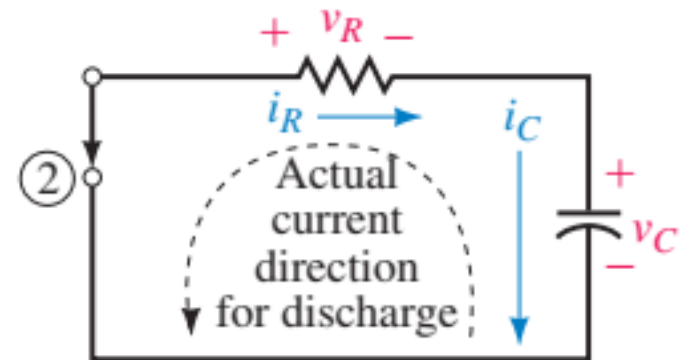


Fig. 24: Discharging case

8. More Complex Circuits (2/15)

EXAMPLE 6

For the circuit of Figure 25(a), determine expressions for v_C and i_C . Capacitors are initially uncharged.

$I = 50 \text{ mA}$, $R_1 = 3 \text{ k}\Omega$, $R_2 = 6 \text{ k}\Omega$, $C_1 = 8 \text{ }\mu\text{F}$, $C_2 = 2 \text{ }\mu\text{F}$.

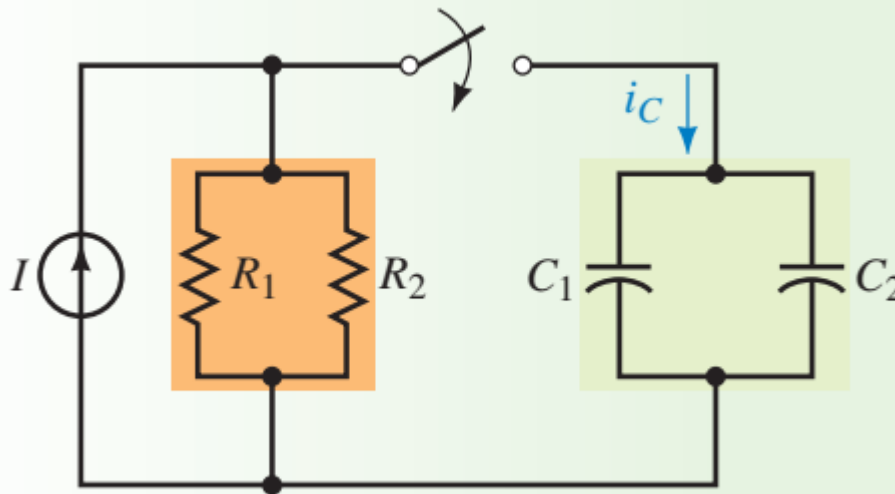


Fig. 25(a)

8. More Complex Circuits (3/15)

Solution $R_{eq} = 3\text{ k}\Omega \parallel 6\text{ k}\Omega = 2\text{ k}\Omega$, $C_{eq} = 8\text{ }\mu\text{F} \parallel 2\text{ }\mu\text{F} = 10\text{ }\mu\text{F}$. The reduced circuit is shown in (b). Converting to a voltage source representation yields (c).

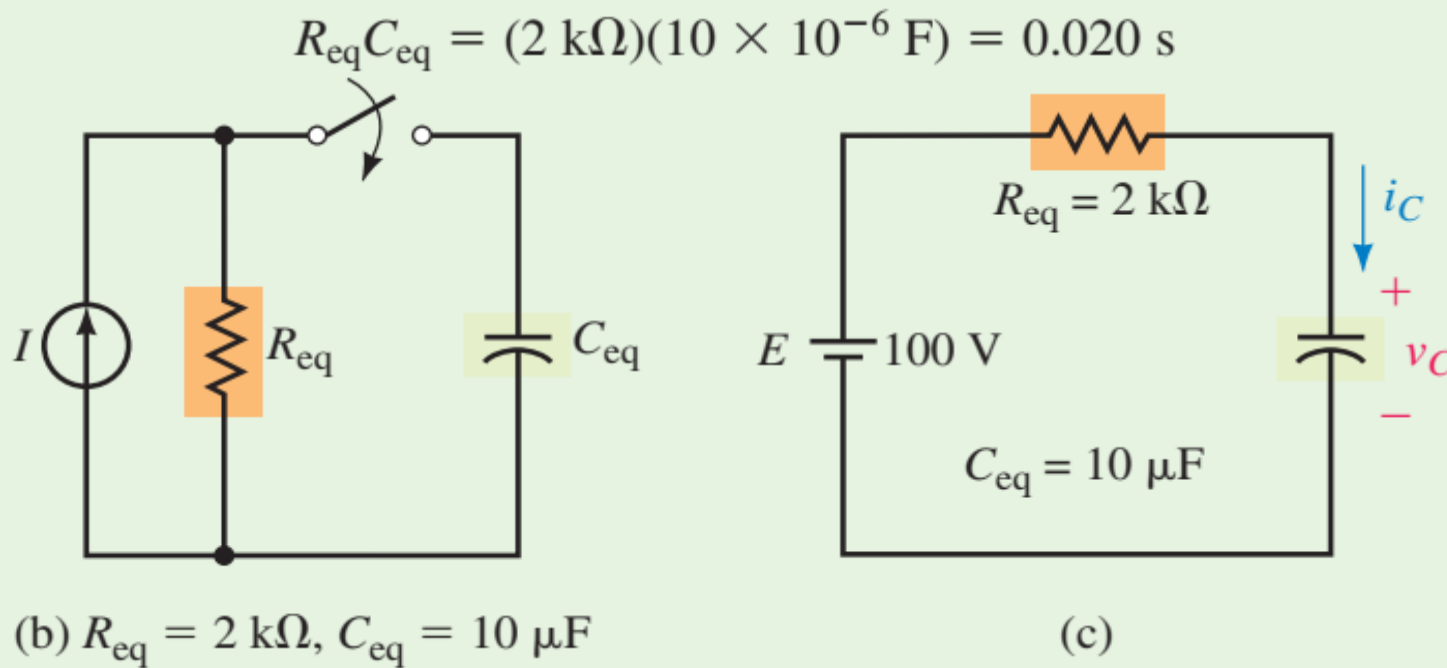


Fig. 25 (b) and (c)

8. More Complex Circuits (4/15)

Thus,

$$v_C = E(1 - e^{-t/R_{\text{eq}}C_{\text{eq}}}) = 100(1 - e^{-t/0.02}) = 100(1 - e^{-50t}) \text{ V}$$

$$i_C = \frac{E}{R_{\text{eq}}} e^{-t/R_{\text{eq}}C_{\text{eq}}} = \frac{100}{2000} e^{-t/0.02} = 50 e^{-50t} \text{ mA}$$

8. More Complex Circuits (5/15)

EXAMPLE 7 The capacitor of Figure 26 is initially uncharged. Close the switch at $t = 0$ s.

- Determine the expression for v_C .
- Determine the expression for i_C .
- Determine capacitor current and voltage at $t = 5$ ms.

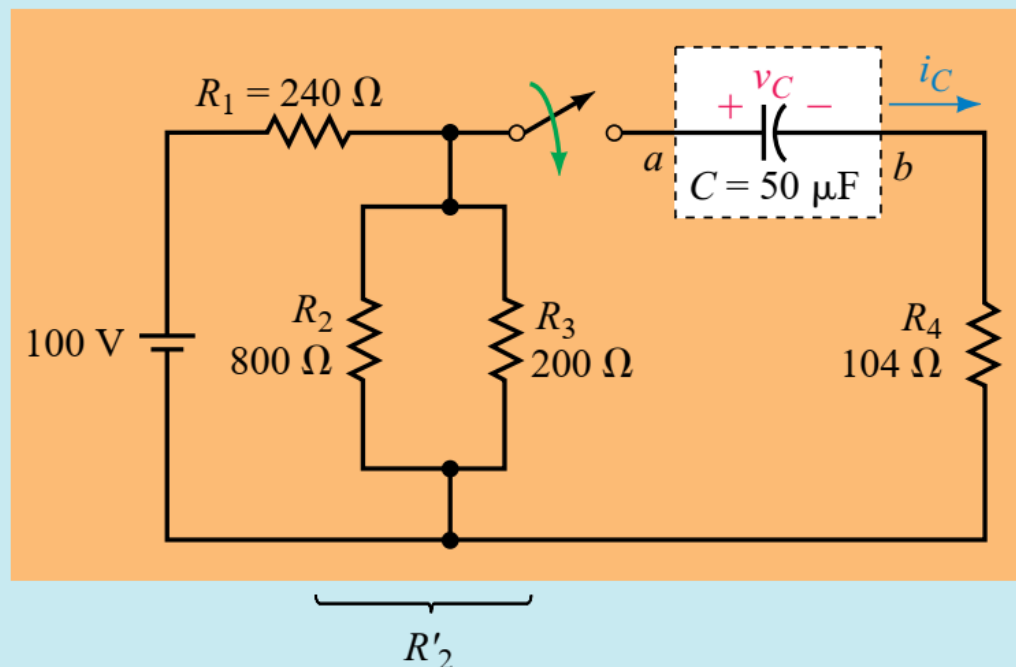


Fig. 26

8. More Complex Circuits (6/15)

Solution Reduce the circuit to its series equivalent using Thévenin's theorem:

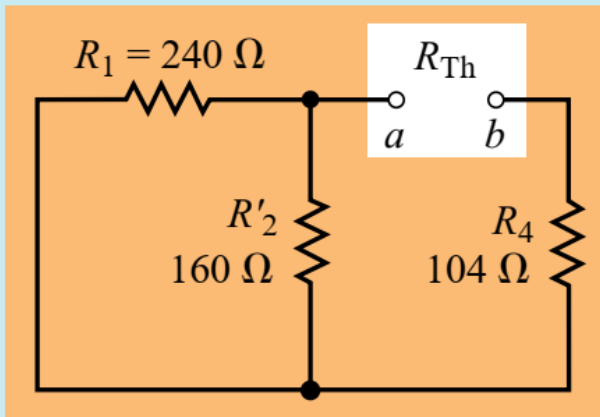
$$R'_2 = R_2 \parallel R_3 = 160 \, \Omega$$

From Figure 27(a) *given below*

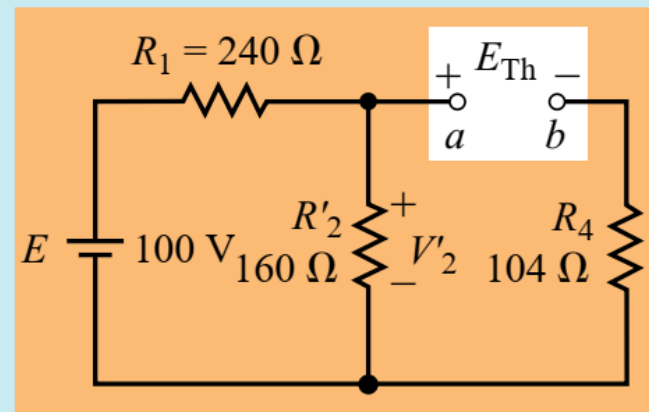
$$R_{Th} = R_1 \parallel R'_2 + R_4 = 240 \parallel 160 + 104 = 96 + 104 = 200 \, \Omega$$

From Figure 27(b) *given below*

$$V'_2 = \left(\frac{R'_2}{R_1 + R'_2} \right) E = \left(\frac{160}{240 + 160} \right) \times 100 \, \text{V} = 40 \, \text{V}$$



(a) Finding R_{Th}



(b) Finding E_{Th}

Fig. 27

8. More Complex Circuits (7/15)

From KVL, $E_{Th} = V'_2 = 40$ V. The resultant equivalent circuit is shown in Figure 28 below:

$$\tau = R_{Th}C = (200\ \Omega)(50\ \mu\text{F}) = 10\ \text{ms}$$

a. $v_C = E_{Th}(1 - e^{-t/\tau}) = 40(1 - e^{-100t})$ V

b. $i_C = \frac{E_{Th}}{R_{Th}}e^{-t/\tau} = \frac{40}{200}e^{-t/0.01} = 200e^{-100t}$ mA

c. $i_C = 200e^{-100(5\ \text{ms})} = 121$ mA. Similarly, $v_C = 15.7$ V

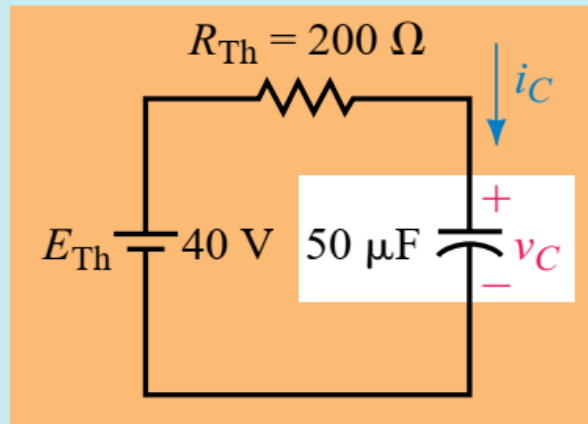


Fig. 28

8. More Complex Circuits (8/15)

- **Important Note:**

- When the switch is moved to its new position, shift the time reference and make $t = 0$ s the time at which the switch is moved to its new position, determine new expressions for V_O and i_{OC} then use continuity considerations to link the two solutions together. (These new expressions are only valid from the new $t = 0$ s reference point. The old expressions are not valid on the new time scale.)
- We now have two time scales for the same graph. However, we generally only show the first scale explicitly; the second scale is implied rather than shown. Computations based on the new equations must use the new scale.

8. More Complex Circuits (9/15)

EXAMPLE 8

The capacitor of Figure 29(a) is uncharged. The switch is moved to position 1 for 10 ms, then to position 2, where it remains.

- Determine v_C during charge.
- Determine i_C during charge.
- Determine v_C during discharge.
- Determine i_C during discharge.
- Sketch the charge and discharge waveforms.

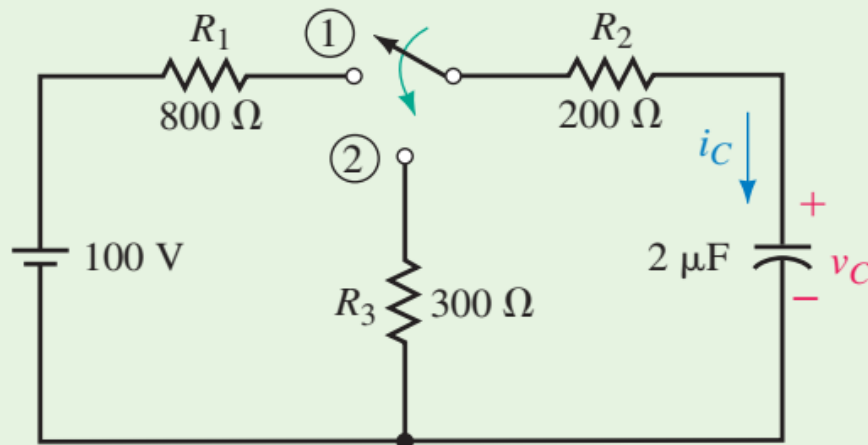


Fig. 29: (a) Full circuit

8. More Complex Circuits (10/15)

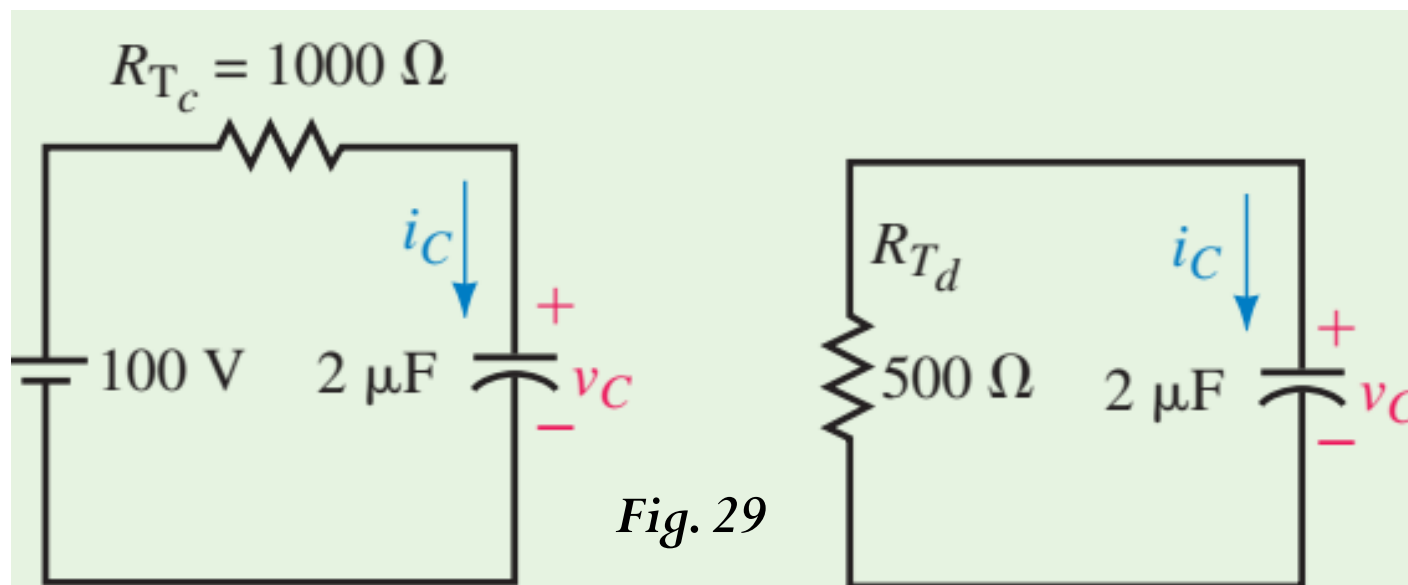


Fig. 29

(b) Charging circuit

$$R_{T_c} = R_1 + R_2$$

(c) Discharging circuit

$$V_0 = 100 \text{ V at } t = 0 \text{ s}$$

R_{T_c} is the total resistance of the charge circuit, while R_{T_d} is the total resistance of the discharge circuit.

8. More Complex Circuits (11/15)

Solution Figure 11–24(b) shows the equivalent charging circuit. Here,

$$\tau_c = (R_1 + R_2)C = (1 \text{ k}\Omega)(2 \text{ }\mu\text{F}) = 2.0 \text{ ms.}$$

a. $v_C = E(1 - e^{-t/\tau_c}) = 100(1 - e^{-500t}) \text{ V}$

b. $i_C = \frac{E}{R_{Tc}} e^{-t/\tau_c} = \frac{100}{1000} e^{-500t} = 100e^{-500t} \text{ mA}$

Since $5\tau_c = 10 \text{ ms}$, charging is complete by the time the switch is moved to discharge. Thus, $V_0 = 100 \text{ V}$ when discharging begins. (This is the continuity linkage.)

c. Figure 11–24(c) shows the equivalent discharge circuit.

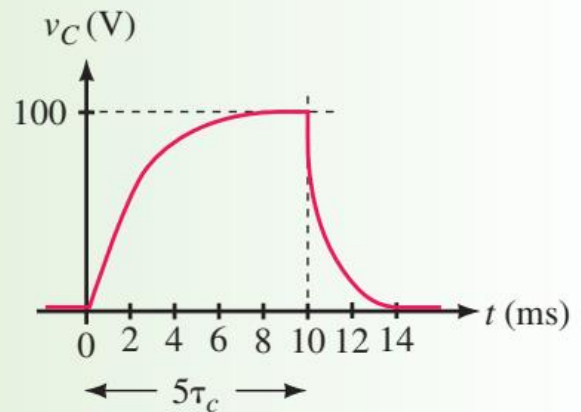
$$\tau_d = (500 \text{ }\Omega)(2 \text{ }\mu\text{F}) = 1.0 \text{ ms}$$

$$v_C = V_0 e^{-t/\tau_d} = 100e^{-1000t} \text{ V}$$

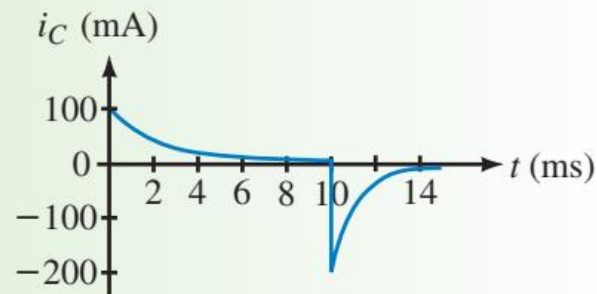
d. $i_C = -\frac{V_0}{R_2 + R_3} e^{-t/\tau_d} = -\frac{100}{500} e^{-1000t} = -200e^{-1000t} \text{ mA}$

8. More Complex Circuits (12/15)

e. See Figure 30. Note that discharge is more rapid than charge since $\tau_d < \tau_c$.



(a)



(b)

Fig. 30

8. More Complex Circuits (13/15)

EXAMPLE 9

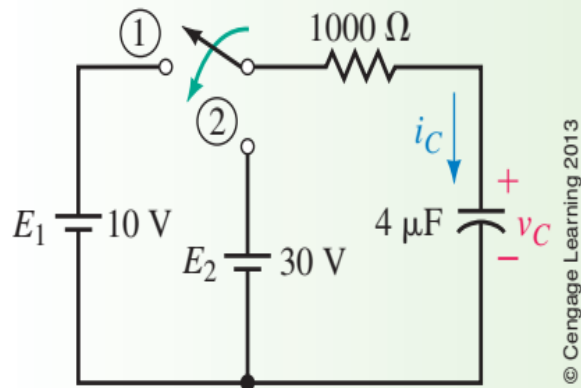


Fig. 31:

The capacitor of Figure 31 is uncharged. The switch is moved to position 1 for 5 ms, then to position 2 and left there.

- Determine v_C while the switch is in position 1.
- Determine i_C while the switch is in position 1.
- Compute v_C and i_C at $t = 5$ ms.
- Determine v_C while the switch is in position 2.
- Determine i_C while the switch is in position 2.
- Sketch the voltage and current waveforms.
- Determine v_C and i_C at $t = 10$ ms.

8. More Complex Circuits (14/15)

Solution

$$\tau_c = \tau_d = RC = (1 \text{ k}\Omega)(4 \text{ }\mu\text{F}) = 4 \text{ ms}$$

a. $v_C = E_1(1 - e^{-t/\tau_c}) = 10(1 - e^{-250t}) \text{ V}$

b. $i_C = \frac{E_1}{R}e^{-t/\tau_c} = \frac{10}{1000}e^{-250t} = 10e^{-250t} \text{ mA}$

c. At $t = 5 \text{ ms}$,

$$v_C = 10(1 - e^{-250 \times 0.005}) = 7.14 \text{ V}$$

$$i_C = 10e^{-250 \times 0.005} \text{ mA} = 2.87 \text{ mA}$$

d. In position 2, $E_2 = 30 \text{ V}$, and $V_0 = 7.14 \text{ V}$. (V_0 is the continuity linkage.)
Use Equation 5:

$$\begin{aligned} v_C &= E_2 + (V_0 - E_2)e^{-t/\tau_d} = 30 + (7.14 - 30)e^{-250t} \\ &= 30 - 22.86e^{-250t} \text{ V} \end{aligned}$$

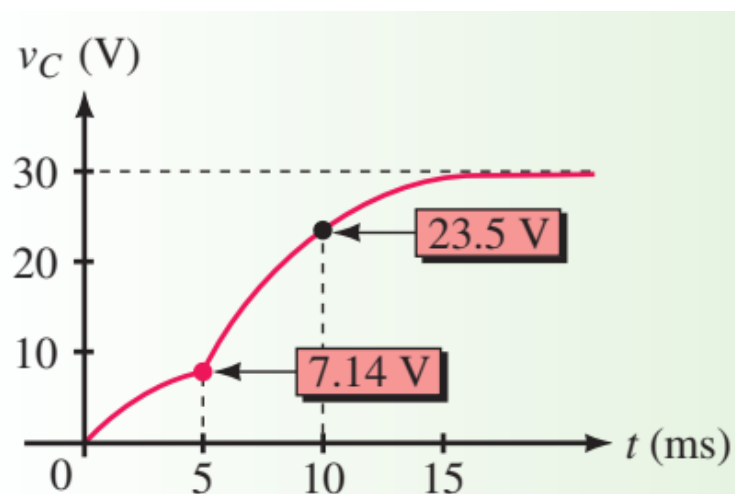
where $t = 0 \text{ s}$ has been redefined for position 2.

8. More Complex Circuits (15/15)

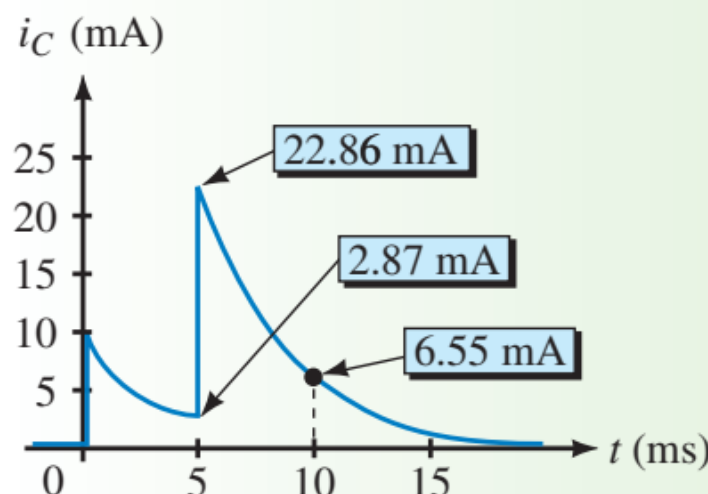
e.
$$i_C = \frac{E_2 - V_0}{R} e^{-t/\tau_d} = \frac{30 - 7.14}{1000} e^{-250t} = 22.86 e^{-250t} \text{ mA}$$

f. See Figure 32.

g. $t = 10 \text{ ms}$ is 5 ms into the new time scale. Thus, $v_C = 30 - 22.86 e^{-250(5 \text{ ms})} = 23.5 \text{ V}$ and $i_C = 22.86 e^{-250(5 \text{ ms})} = 6.55 \text{ mA}$. Values are plotted on the graph.



(a)



(b)

Fig. 32:

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*